# Chapter 13. Mastery Learning

***Summary:*** *By organizing its curriculum into a knowledge graph that keeps track of prerequisite relationships between topics, Math Academy is able to implement mastery learning, a strategy in which students demonstrate proficiency on prerequisites before advancing. While even loose approximations of mastery learning have been shown to produce massive gains in student learning, mastery learning faces limited adoption due to clashing with traditional teaching methods and placing increased demands on educators. Math Academy implements true mastery learning at a fully granular level, which requires fully individualized instruction and is only attainable through one-on-one tutoring.*

## Mastery Learning is Underused

One of our main paradigms is **mastery learning**, also proposed by Bloom (1968), in which students must demonstrate proficiency on prerequisite topics before moving on to more advanced topics. True mastery learning at a fully granular level requires fully individualized instruction, which is only attainable through one-on-one tutoring.

There are methods by which a single teacher can loosely approximate mastery learning, such as Bloom's Learning For Mastery (LFM) strategy and Keller's Personalized System of Instruction (PSI). As Kulik, Kulik, & Bangert-Drowns (1990) summarize:

> *"In both LFM and PSI courses, material to be learned is divided into short units, and students take formative tests on each unit of material (Bloom, 1968; Keller, 1968). ... Lessons in LFM courses are teacher presented, and students move through these courses at a uniform, teacher-controlled pace. Lessons in PSI courses are presented largely through written materials, and students move through these lessons at their own rates."*

However, as Bloom (1984) discovered when characterizing the two-sigma problem, a single teacher practicing mastery learning with 30 students could only produce a one-sigma effect size as compared to the two-sigma effect size of individual tutoring. And while numerous studies reproduced the finding that even loose approximations of mastery learning (managed manually by a single teacher) produce substantial learning gains, most studies were unable to reproduce gains as strong as one sigma (the average effect size was about 0.5 standard deviations) (Kulik, Kulik, & Bangert-Drowns, 1990):

> *"The data show that mastery learning programs have positive effects on student achievement. On the average, such programs raise final examination scores by about 0.5 standard deviations, or from the 50th to the 70th percentile, in colleges, high schools, and the upper grades of elementary schools. Although PSI and LFM strategies differ on several points and the two teaching methods have been studied in distinct ways, studies of PSI and LFM report similar results. PSI raised examination scores by an average of 0.48 standard deviations; LFM raised examination scores by an average of 0.59 standard deviations."*

Unfortunately, despite producing well-documented learning gains in classrooms, even loose approximations of mastery learning were not widely adopted as they faced opposition for deviating from traditional convention and requiring more effort from teachers and administrators (Sherman, 1992). (It's true that a minority of teachers now attempt some degree of differentiated instruction, but this is not the same as true mastery learning, which holds all students to the same standard and is completely individualized.)

As lamented by John Gilmour Sherman (1992), who was a co-creator, researcher, and practitioner of Keller's Personalized System of Instruction (PSI):

> *"Some PSI courses have been prohibited in spite of their success. I know of several colleagues who were given 'cease and desist' orders. Some are names prominent in the literature, their courses effective, according to objective data.*
>
> *I experienced this also. Avoiding a frontal attack, the chairman of the Psychology Department at Georgetown declared by fiat that something on the order of 50% of class time must be devoted to lecturing. By reducing the possibility of self-pacing to zero, this effectively eliminated PSI courses.*
>
> *He issued this order on the grounds that in the context of lecturing 'it is the dash of intellects in the classroom that informs the student.' No data were presented on this point! The spectacle of purporting to defend scholarship while deciding the merits of instructional methods by assertion is silly.*
>
> *The troubling aspect of all these cases was that data played no part in the decisions. It is disturbing when one has to wonder whether research on the education process makes any difference."*

As Buskist, Cush, & DeGrandpre (1991) elaborate, mastery learning methods like PSI were shot down because they threatened the traditional educational establishment:

> *"The first and most important task of any institution is self-preservation. Once in place, its primary goal must be to hold its ground or, if possible, advance. If this goal is not met, then its demise is eminent. PSI poses certain implications that threaten the preservation of the educational establishment and its guardians (department heads, deans, and academic vice-presidents). According to Keller,*
>
> *'Suppose that a system such as PSI were to be given official approval for adoption throughout the educational scale from top to bottom... With every student given individual treatment, when would formal education start?... What would happen to the classroom hour, the college quarter,*

*the semester, or the academic year?... Who would win the scholarships and prizes? Who would make Phi Beta Kappa? Who would be the class valedictorian?... What changes would be made in the payment of tuition when the period of course attendance varied? How would a course of study be defined?'*

*In other words, the major impediment to educational reform is the educational system itself. That is perhaps why all major efforts at educational reform in this century have been directed at renovating curricula and not at changing how teachers teach (see, e.g., Skinner, 1984; McGovern, 1990).*

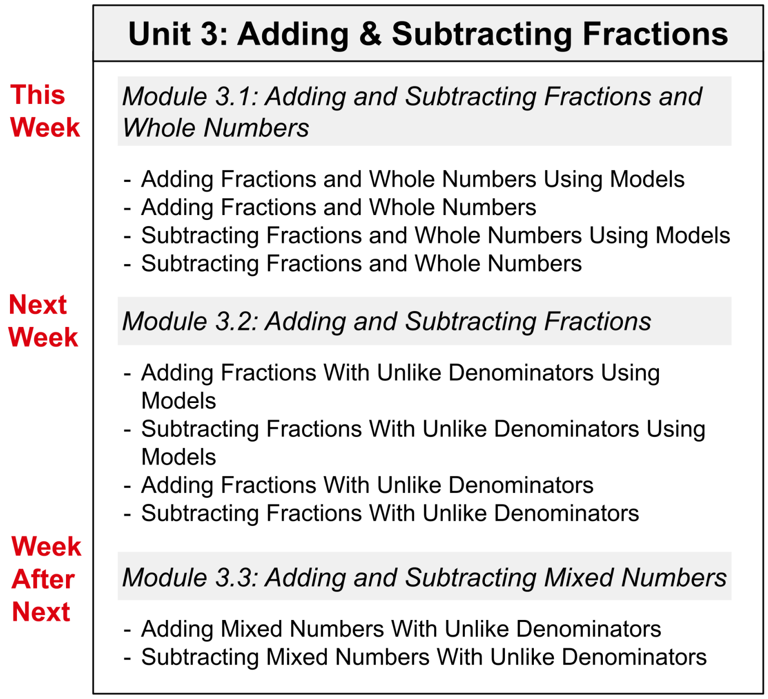
*Revamping curricula requires no revamping of the educational establishment. The curriculum changes, but that is all. Courses are still taught by lecture within a term's time. Grade distributions still approximate the normal curve and students enroll in upper division courses without first mastering more fundamental material. Students still take about four years, give or take a term, to finish what higher education demands of them.*

*The Keller Plan runs contrary to this strategy. It is a bold attempt to change how we teach, despite what we teach. In an entirely PSI-based college, students might finish in two years, maybe sooner, and learn a good deal more. Imagine what would happen if the entire educational system were PSI-based: huge numbers of people, most still in the throes of puberty, might be graduating from college-an unsettling thought for many educators. Indeed, PSI represents a threat to the educational system and its guardians.*

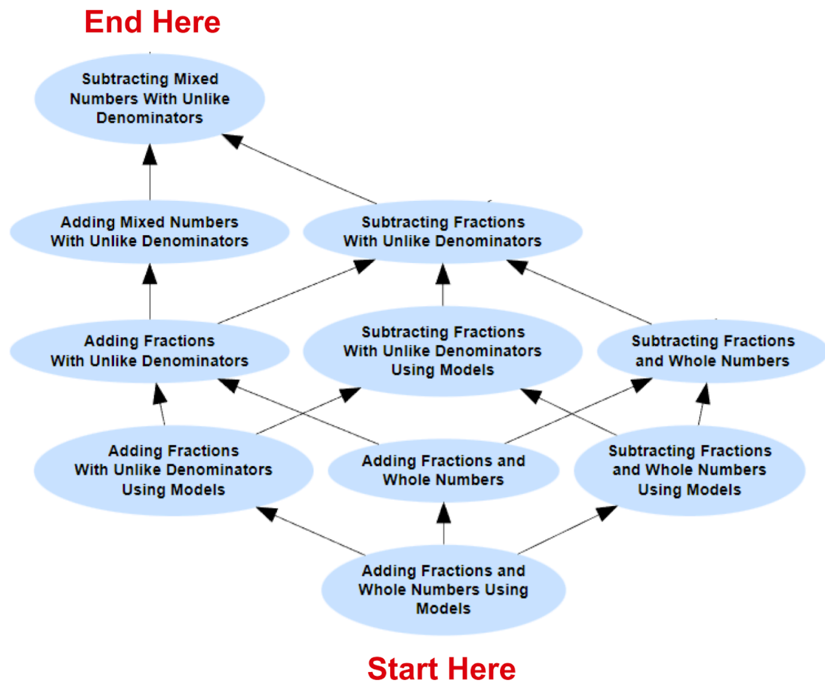
*PSI simply does not fit well into our modern educational system. The longstanding tradition of teaching by lecture has accumulated inertia that has proven difficult to dislodge. In the interests of self-preservation, the educational establishment has backed reforms favoring what teachers teach instead of how teachers teach."*

## Implementing Mastery Learning

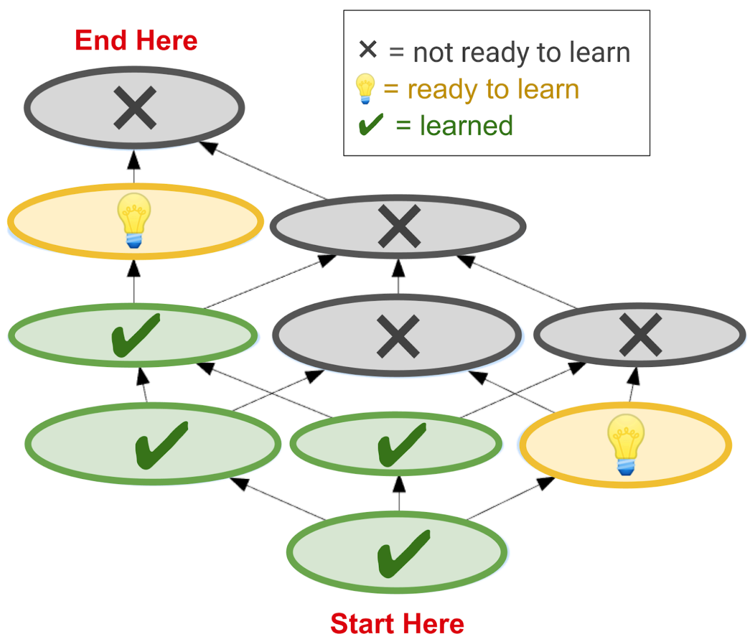
The traditional convention is to march students through a linear sequence of topics according to a predetermined schedule (like that shown below). Any students who get lost are continually asked to learn new topics despite not having mastered the prerequisites. As a result, those students spend class time learning little to nothing and developing a general distaste for math.




Math Academy, however, implements true mastery learning at a fully granular level. We accomplish this by organizing topics into a **knowledge graph** that shows all the topics and **prerequisite relationships** between them. In the knowledge graph below, arrows point from simpler prerequisite topics to more advanced "post"-requisite topics.




By overlaying a student's progress on the knowledge graph, we can identify the topics that they are ready to learn – that is, the topics for which they have demonstrated a sufficient level of proficiency on the prerequisites. We only serve students lessons on these topics. If a student gets stuck on a topic, they can try again another day, but in the meantime, they are allowed to learn other topics that don't depend on the problematic one.



It is infeasible for a single teacher, who can only teach one topic at a time, to manually support true mastery learning across a class full of students who all have different learning profiles. As researchers have discovered, knowledge profiles vary immensely even across students in the same grade (Pedersen, et al., 2023):

> *"our results suggest that nearly 38% and 49% of students in grade four and eight classrooms may either struggle to understand 'grade-level' content or have already mastered the content, respectively."*

However, Math Academy's fully-automated interactive lessons support de-synchronized learning where different students are simultaneously taught different topics. With a software system that emulates the decisions of an expert tutor, we are able to provide fully individualized instruction at scale and achieve true mastery learning at a fully granular level.

## Knowledge Frontier as Zone of Proximal Development

Mastery learning is closely related to Vygotsky's **Zone of Proximal Development**, which refers to the range of tasks that a student is able to perform while supported, but cannot do on their own. Students maximize their learning when they are completing tasks within this range.

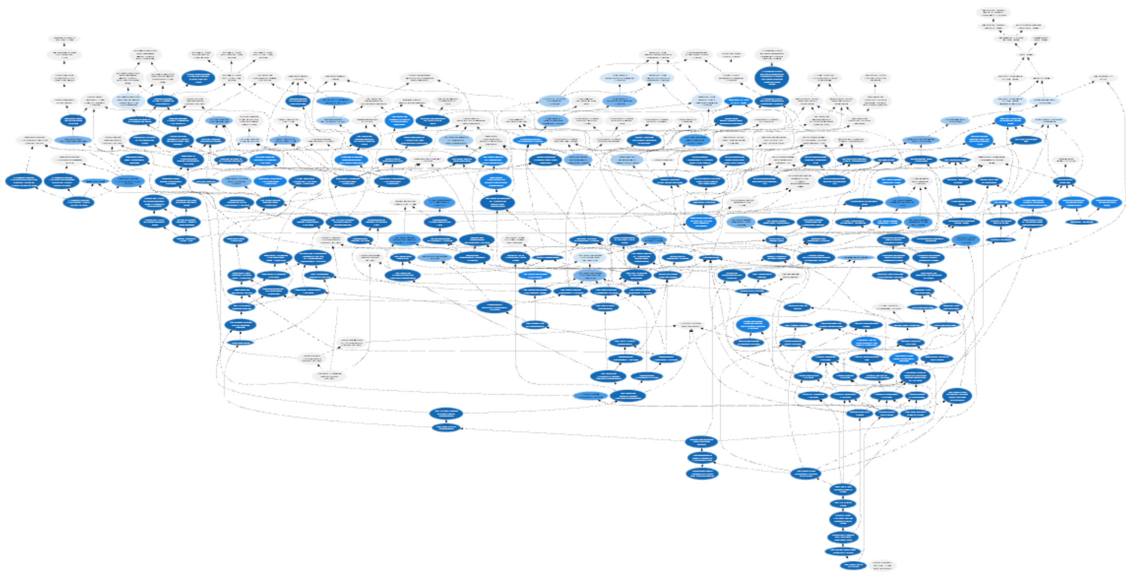
In the context of an adaptive learning system, a student's zone of proximal development coincides with their **knowledge frontier** or **edge of mastery**, the set of new topics for which they have mastered the prerequisites. Selecting new learning tasks from a student's knowledge frontier can lead to drastic improvements in learning.

For instance, another learning platform has reported (Zou et al., 2019) that even when a teacher has access to student performance data chooses a new topic that they believe is appropriate for the class, a student is about 3-4x as likely to be successful in mastering that topic if it lies along their knowledge frontier (as opposed to residing beyond their frontier). This led the authors to conclude that

> *"...[S]imply providing teachers with data is not always sufficient for good instructional decision making. ... Optimizing learning outcomes requires correct teaching decisions that lead students on the right path, based on the student's ZPD [Zone of Proximal Development]."*

A student's knowledge frontier can be visualized as the edge of their **knowledge profile**, which, loosely speaking, represents how "developed" their mathematical brain is. Every time they learn a new math topic, it's as if they grow a new brain cell and connect it to existing brain cells. Initially, this new brain cell is weak and requires frequent nurturing, but over time it becomes strong and requires less frequent care.

For instance, a knowledge profile for a second-semester calculus student is visualized below. Learned topics are shaded (with darker shading indicating that more successful practice has been completed), and arrows between topics represent prerequisite relationships.

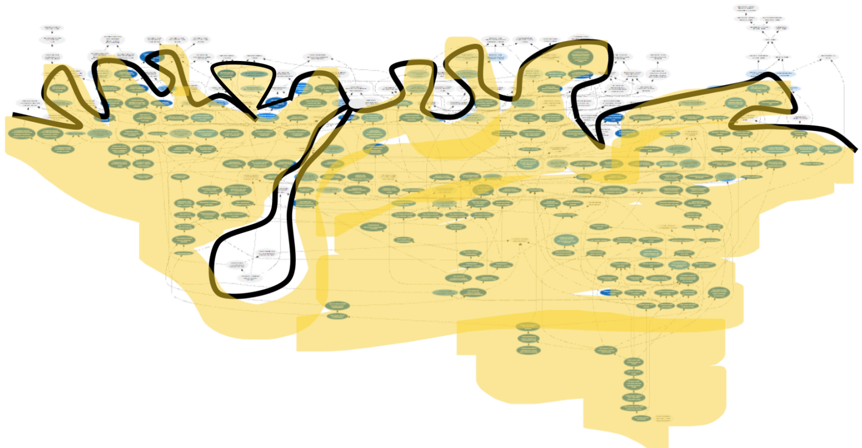


(Note that this visualization only shows a "subsystem" within the student's full mathematical brain – there are several hundred topics in the calculus course, but there are thousands of topics in Math Academy's entire mathematical curriculum, which spans elementary school through university-level math.)

The **knowledge frontier** is the edge of the knowledge profile. It separates what the student knows from what they don't know.

- The student knows all the simpler topics below their knowledge frontier. That is, they know all the prerequisites, the prerequisites of the prerequisites, and so on.

- However, they do not know any of the advanced topics at or above their knowledge frontier.



When a student first starts on the Math Academy system, we begin with a **placement diagnostic**, a **dynamic assessment** that quickly estimates their knowledge frontier. Following the diagnostic, whenever a student is served new lessons, those lessons always cover topics that are on the student's estimated knowledge frontier, and the estimated knowledge frontier quickly becomes more accurate as the student completes those lessons.

# Key Papers

***Note:*** *"Importance" blurbs may include pieces of direct quotes referenced earlier in this chapter. If citing this chapter, cite from the body (above).*

- Bloom, B. S. (1984). The 2 sigma problem: The search for methods of group instruction as effective as one-to-one tutoring. *Educational researcher*, 13(6), 4-16.

  ***Importance:*** *The average tutored student performed better than 98% of the students in a traditional class, an effect size of two sigmas (standard deviations). However, a single teacher practicing mastery learning with 30 students could only produce a one-sigma effect size as compared to the two-sigma effect size of individual tutoring.*

- Kulik, C. L. C., Kulik, J. A., & Bangert-Drowns, R. L. (1990). Effectiveness of mastery learning programs: A meta-analysis. *Review of educational research*, 60(2), 265-299.

  ***Importance:*** *Numerous studies reproduced the finding that even loose approximations of mastery learning (managed manually by a single teacher) produce substantial learning gains, though generally not as high as one sigma (the average effect size was about 0.5 standard deviations).*

- Sherman, J. G. (1992). Reflections on PSI: Good news and bad. *Journal of Applied Behavior Analysis*, 25(1), 59.

  Buskist, W., Cush, D., & DeGrandpre, R. J. (1991). The life and times of PSI. *Journal of Behavioral Education, 1*, 215-234.

  ***Importance:*** *Despite producing well-documented learning gains in classrooms, Keller's Personalized System of Instruction (a method by which a single teacher can loosely approximate mastery learning) was not widely adopted as it faced opposition for deviating from traditional convention and requiring more effort from teachers and administrators.*

- Pedersen, B., Makel, M. C., Rambo-Hernandez, K. E., Peters, S. J., & Plucker, J. (2023). Most mathematics classrooms contain wide-ranging achievement levels. *Gifted Child Quarterly,*

*67*(3), 220-234.

***Importance:*** *Knowledge profiles vary immensely even across students in the same grade: nearly 38% and 49% of students in grade four and eight classrooms may either struggle to understand "grade-level" content or have already mastered the content, respectively.*

- Zou, X., Ma, W., Ma, Z., & Baker, R. S. (2019). Towards helping teachers select optimal content for students. In *Artificial Intelligence in Education: 20th International Conference, AIED 2019, Chicago, IL, USA, June 25-29, 2019, Proceedings, Part II 20* (pp. 413-417). Springer International Publishing.

  ***Importance:*** *Another learning platform has reported that even when a teacher has access to student performance data and chooses a new topic that they believe is appropriate for the class, a student is about 3-4x as likely to be successful in mastering that topic if it lies along their knowledge frontier (as opposed to residing beyond the frontier).*